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Studierichting: Informatica
Oefeningenreeks 5B nr. 2.4

$$\begin{aligned} & \int_{-\infty}^{+\infty} x e^{-x^2} dx \\ &= \lim_{a \rightarrow -\infty} \int_a^0 x e^{-x^2} dx + \lim_{b \rightarrow +\infty} \int_0^b x e^{-x^2} dx \end{aligned}$$

Eerste limiet:

$$\begin{aligned} & \lim_{a \rightarrow -\infty} \int_a^0 x e^{-x^2} dx \\ &= \lim_{a \rightarrow -\infty} \left[-\frac{1}{2} e^{-x^2} \right]_a^0 \\ &= \lim_{a \rightarrow -\infty} \left(-\frac{1}{2} e^0 + \frac{1}{2} e^{-a^2} \right) \\ &= \lim_{a \rightarrow -\infty} \left(-\frac{1}{2} + \frac{1}{2} e^{-a^2} \right) \\ &= -\frac{1}{2} \end{aligned}$$

Tweede limiet:

$$\begin{aligned} & \lim_{b \rightarrow +\infty} \int_0^b x e^{-x^2} dx \\ &= \lim_{b \rightarrow +\infty} \left[-\frac{1}{2} e^{-x^2} \right]_0^b \\ &= \lim_{b \rightarrow +\infty} \left(-\frac{1}{2} e^{-b^2} + \frac{1}{2} \right) \\ &= \frac{1}{2} \end{aligned}$$

Dus:

$$\begin{aligned} & -\frac{1}{2} + \frac{1}{2} = 0 \\ & \Rightarrow \int_{-\infty}^{+\infty} x e^{-x^2} dx = 0 \end{aligned}$$

References